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of water is xylem. Phloem is not colored indicating that it has no role in the ascent of sap (Figure 11.12).

Mechanism of Ascent of Sap

In ascent of sap, the biggest challenge is the force required to lift the water to the top of the tallest trees. A number of theories have been put forward to explain the mechanism of the ascent of sap. They are, A. Vital force theories, B. Root pressure theory, and C. Physical force theory.

11.5.2 Vital Force Theories

According to vital force theories, living cells are mandatory for the ascent of sap. Based on this the following two theories derived:

1. **Relay pump theory of Godlewski (1884)**

Periodic changes in osmotic pressure of living cells of the xylem parenchyma and medullary ray act as a pump for the movement of water.

2. **Pulsation theory of J.C.Bose (1923)**

Bose invented an instrument called **Crescograph**, which consists of an electric probe connected to a galvanometer (Figure 11.13). When a probe is inserted into the inner cortex of the stem, the galvanometer showed high electrical activity. Bose believed a rhythmic pulsating movement of inner cortex like a pump (similar to the beating of the heart) is responsible for the ascent of sap. He concluded that cells associated with xylem exhibit pumping action and pumps the sap laterally into xylem cells.



Figure 11.13: J.C. Bose

Objections to vital force theories

i. **Strasburger** (1889) and **Overton** (1911) experimentally proved that living cells are not mandatory for the ascent of sap. For this, he selected an old oak tree trunk which when immersed in **picric acid** and subjected to excessive heat killed all the living cells of the trunk. The trunk when dipped in water, the ascent of sap took place.

ii. Pumping action of living cells should be in between two xylem elements (vertically) and not on lateral sides.

11.5.3 Root Pressure Theory

If a plant which is watered well is cut a few inches above the ground level, sap exudes out with some force. This is called **sap exudation** or **bleeding**. **Stephen Hales**, father of plant physiology observed this phenomenon and coined the term '**Root Pressure'**. **Stoking** (1956) defined root pressure as "*a pressure developing in the tracheary elements of the xylem as a result of metabolic activities of the root*". But the following objections have been raised against root pressure theory:

i. Root pressure is totally absent in gymnosperms, which includes some of the tallest plants.

ii. There is no relationship between the ascent of sap and root pressure. For